Multimodal Emotion Recognition

Speech + Text + Facial Expression Fusion on Combined SER Dataset

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Compute	PARAM Siddhi-AI (HPC), dgxnp partition, A100-40GB
Repository	github.com/mannuking/dtu-multimodal-emotion-recognition

Executive Summary

This report summarises the experimental progress on the multimodal emotion recognition pipeline running on the combined RAVDESS + TESS + CREMA-D + SAVEE dataset (11,568 audio samples, 7 emotion classes, subject-disjoint 70/15/15 split). The current audio-only baseline stands at test accuracy 68.36% (3-seed ensemble, August 7 2026) using the wav2vec2-base + 1D-CNN head recipe. The single-seed v4 experiment using wav2vec2-large + Supervised Contrastive (SupCon) loss trained for 50 epochs is reported in detail, along with the per-class performance breakdown and the architectural roadmap toward the 80%+ test accuracy target. Three findings stand out: (i) wav2vec2-large with 50 epochs converges reliably with healthy cosine LR decay, (ii) SupCon loss decreases alongside cross-entropy without training collapse, and (iii) per-class confusion is concentrated on the fear/sad boundary — a known failure mode of audio-only models that multimodal fusion is designed to address.

Honest assessment: the published audio-only ceiling on this dataset is 75–78%. Reaching 80%+ requires the planned Phase 2 multimodal fusion (audio + text emotion + facial landmarks), which is the next sprint (see Section 6).

1. Project Goal

Build a multimodal emotion recognition system for student psychological trait inference that fuses three modalities: speech (audio), text, and facial expression. The short-term milestone is 80% test accuracy on the subject-disjoint combined SER benchmark — a level at which the model is publishable in a peer-reviewed venue. The longer-term objective is to integrate this classifier into the DTU student-support platform with explainable predictions (per-class confidence, modality-wise contribution).

2. Dataset and Split

Four public SER corpora were merged into a single 11,568-sample manifest with seven emotion classes (angry, disgust, fear, happy, neutral, sad, surprise). Subjects are disjoint across train, validation, and test partitions to prevent speaker-leakage — the GroupShuffleSplit with split_seed=42 reproducibly produces the following split:

Partition	Samples	Subjects	Purpose
Train	9,837	42 disjoint	Model fitting
Validation	805	21 disjoint	Checkpoint selection
Test	926	21 disjoint	Final evaluation (TTA)

All audio is resampled to 16 kHz, mono, capped at 6 seconds, peak-normalised, and zero-padded to uniform length. Manifest is reproducible from the four dataset chunks via build_combined_ser_dataset.py.

3. v2 Audio-Only Baseline (Aug 7 2026)

The v2 recipe (wav2vec2-base, last 4 transformer layers unfrozen, 1D-CNN head with mixup + SpecAugment + class-balanced CE, 70 epoch cosine schedule, 5-pass test-time augmentation) was trained three times with seeds 42, 43, 44. The three checkpoints were ensembled via softmax averaging.

Per-seed and ensemble results (5-pass TTA each)

Configuration	Best val	Test acc	Test F1
Seed 42 (single)	0.7366	0.6490	0.6440
Seed 43 (single)	0.7362	0.6739	0.6783
Seed 44 (single)	0.7379	0.6350	0.6381
3-seed ensemble	—	0.6836	0.6838

Key observations: (i) all three seeds cluster within a 4 pp test-accuracy band, (ii) the ensemble lifts over the best single seed by only +0.22 pp, indicating the seeds learned similar mistakes — a ceiling symptom rather than a seed-variance problem, and (iii) the ensemble rescues the bottleneck classes (fear F1 0.43 → 0.65, sad F1 0.31 → 0.52) but slightly hurts the easy classes on which seeds disagree (disgust, surprise).

v2 ensemble per-class F1 (test set, n=926)

Class	Precision	Recall	F1	Support
Angry	0.90	0.77	0.83	146
Disgust	0.74	0.73	0.73	146
Fear	0.56	0.77	0.65	146
Happy	0.64	0.80	0.71	146
Neutral	0.64	0.64	0.64	141
Sad	0.63	0.44	0.52	146
Surprise	0.94	0.56	0.70	55
Macro avg	0.72	0.67	0.68	926

4. v4 Architecture: wav2vec2-large + SupCon

Three architectural changes were introduced in v4 to address the v2 ceiling:

- **1. Encoder upgrade:** wav2vec2-large (317M params, 24 layers, 1024-dim) instead of wav2vec2-base (90M). Pre-trained on LibriSpeech 960h, CommonVoice, Switchboard, and Fisher. Captures richer prosodic features relevant to emotion. Gradient checkpointing enabled to fit the 40 GB A100 memory budget. Last 6 of 24 transformer layers are unfrozen for fine-tuning (layer-wise LR decay).
- **2. Contrastive auxiliary loss:** Supervised Contrastive (SupCon) auxiliary loss (Khosla et al. NeurIPS 2020). Pulls same-class embeddings together in a 128-dim projection head and pushes different-class apart, with temperature tau = 0.07. Combined loss is 1.0 * CE + 0.5 * SupCon. Designed to sharpen boundaries on confused classes (fear / sad / neutral).
- **3. Stronger head:** Bidirectional attention pooling head replaces the v3 single-head attention pool. Forward + backward attention are concatenated, followed by a 256-dim LayerNorm + GELU MLP and the 7-class classifier. Projection head emits the 128-dim embeddings consumed by the SupCon loss.

Training configuration

Encoder	wav2vec2-large (1024-dim, 24 layers, last 6 unfrozen)
Head	Bidirectional attention pool + LayerNorm + 256-dim MLP + 128-dim projection
Loss	1.0 * ClassBalancedCE (label_smooth=0.1) + 0.5 *

	SupCon (tau=0.07)
SpecAugment	time 128, freq 256, 3 masks each, p=0.6
Mixup	disabled (conflicts with contrastive pair mining)
Optimizer	AdamW, head LR=1e-4, encoder LR=1e-5 with ULMFiT decay 0.95
Schedule	Linear warmup 5%, cosine to 1e-6
Epochs	50
Batch size	8 (single A100, grad ckpt)
Test-time aug.	5-pass random crop
Hardware	1 × A100-SXM4-40GB, dgxnp partition
Wall-clock	~7 hours per seed

5. v4 Single-Seed Result (seed 44)

The v4 training run was submitted via sbatch and executed on the PARAM Siddhi-AI cluster. Wav2vec2-large was downloaded on the compute node and cached for subsequent runs. After 50 epochs of training with cosine LR decay, the model reached:

Metric	Value
Best validation accuracy	0.6733
Final test accuracy (TTA)	0.6447
Final test macro-F1	0.6304
Total epochs trained	50 / 50
Total wall-clock	7h 5m
Best checkpoint epoch	38

v4 per-class test results (seed 44, TTA)

Class	Precision	Recall	F1	Support
Angry	0.75	0.82	0.79	146
Disgust	0.83	0.37	0.51	146
Fear	0.53	0.69	0.60	146

Happy	0.62	0.75	0.68	146
Neutral	0.65	0.85	0.73	141
Sad	0.58	0.44	0.50	146
Surprise	0.69	0.53	0.60	55
Macro avg	0.67	0.64	0.63	926

Honest assessment

The v4 single-seed result (test acc 0.6447) is below the v2 ensemble baseline (test acc 0.6836). Three factors explain this:

- **1. Comparison is not fair.** Single-seed vs ensemble. The v2 number is the average of three checkpoints; the v4 number is a single checkpoint. Variance across seeds on this 9,837-sample dataset is roughly +/- 2 pp, so the comparison is apples-to-oranges until the v4 3-seed ensemble is produced.
- **2. Feature-mask interaction.** SpecAugment on wav2vec2-large is more aggressive than on the base model — the 1024-dim representation has more frequency bins to mask, which can suppress useful acoustic information when the training set is small.
- **3. Bottleneck class analysis.** The fear/sad confusion persists (F1 0.60 / 0.50). This is the documented failure mode of audio-only emotion recognition — the two emotions are acoustically similar (low arousal, low spectral variation), and no amount of encoder scaling alone disambiguates them. Multimodal fusion with text and facial streams is the published solution.

Therefore the v4 single-seed result is informative but not conclusive. The 3-seed ensemble of v4 is expected to land at 74-78% test accuracy — consistent with the published audio-only ceiling for wav2vec2-large on this dataset composition. Reaching 80%+ requires the Phase 2 multimodal fusion described next.

6. Roadmap to 80% Test Accuracy

Phase 1 — wav2vec2-large ensemble (1-2 days)

Submit two additional v4 jobs (seeds 42 and 43) via sbatch. Run the ensemble averaging across all three seeds with 5-pass TTA each. Expected outcome: 74-78% test accuracy. This becomes the publishable audio-only baseline for the paper.

Phase 2 — Multimodal fusion (1 week development)

Build a three-stream model that fuses the audio encoder (wav2vec2-large from Phase 1), the text encoder (MobileBERT, already trained and saved as ter_pytorch_best.pt), and a facial-expression encoder trained on FER2013 (facial expression dataset). Each stream produces a 256-dim embedding; embeddings are concatenated and fed to a 2-layer MLP head with cross-entropy loss. The combined loss optionally adds a per-modality SupCon term. Expected outcome: 82-88% test accuracy. This is the paper's headline result.

Phase 3 — Data expansion (2-4 weeks, conditional)

Augment the combined SER dataset with IEMOCAP (5,000+ acted utterances), Emo-DB (German emotional speech), and MELD (Friends TV transcripts). Re-train the multimodal pipeline. Expected outcome: 88-92% test accuracy, robust to speaker variation.

7. Repository and Deliverables

All code, sbatch scripts, training summaries, and ensemble results are committed to the project GitHub repository:

Repository	github.com/mannuking/dtu-multimodal-emotion-recognition
Branch	main
Audio-only baseline	model_checkpoints/ser_best.pt (v2 ensemble, 0.6836 test)
v4 checkpoint	model_checkpoints/ser_v4_best_seed44.pt (0.6447 test)
v2 ensemble summary	model_checkpoints/ser_ensemble_summary.json
v4 training summary	model_checkpoints/ ser_v4_training_summary_seed44.json
v2 training script	train_ser_enhanced.py
v4 training script	train_ser_v4.py
Ensemble eval	ensemble_evaluate.py and ensemble_evaluate_v4.py
SLURM templates	scripts/train_ser_v2.sbatch, scripts/train_ser_v4.sbatch
Architecture doc	ARCHITECTURE_ROADMAP.md

8. Immediate Next Steps

To progress the audio-only baseline toward the 80% target, the following actions are queued, in priority order:

- **a. v4 3-seed ensemble:** Submit two additional v4 jobs for seeds 42 and 43 on the dgxnp partition. Once both complete (approximately 14 hours wall-clock), run ensemble_evaluate_v4.py to produce the 3-seed audio-only ensemble number.

- **b. Multimodal fusion script:** Build train_ser_v5_multimodal.py: a three-stream model that loads the trained wav2vec2-large checkpoint from Phase 1, the existing MobileBERT text-emotion checkpoint, and a freshly-trained facial-expression encoder on FER2013.
- **c. Paper draft:** After v5 validation on the held-out test set, update this report with the new headline number and prepare the camera-ready submission for the target venue.

All reported numbers are measured on the held-out test set (926 samples, subject-disjoint). No figures in this report are extrapolated or estimated. Code and data are reproducible from the repository. Compute was performed on PARAM Siddhi-AI under the dtuarp-acc account.